TIMOTHY WEAVER

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SUMMARY

Software engineer with two live production SaaS products, built and run solo: a video editing platform with about 3,000 sign-ups on Spring Boot, PostgreSQL, and React with iOS and Android apps, and a multi-tenant event platform on Next.js and TypeScript. Finishing a Master of Computer Science at Arizona State (December 2026) after a B.S. in Software Engineering, summa cum laude.

EDUCATION

Master of Computer Science — Arizona State University Expected Dec 2026

GPA 3.79. Coursework: Foundations of Algorithms, Distributed & Multiprocess Operating Systems, Software Security, Information Assurance & Security, Artificial Intelligence, Statistical Machine Learning, Data Mining

B.S. Software Engineering — Arizona State University May 2025

GPA 3.83, summa cum laude; Dean's List (4 terms)

Coursework: Data Structures & Algorithms, Foundations of Machine Learning, Database Management, Web-Based Applications, Mobile Systems, Secure Software Engineering

SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL

Frameworks: Spring Boot, Next.js, React, Node.js, Capacitor

Data & Infra: PostgreSQL, Redis, Prisma, MongoDB, Docker, AWS S3, Cloudflare R2, GitHub Actions CI/CD, Linux/systemd

Product Eng.: REST API design, JWT/OAuth2, Stripe billing, multi-tenancy, iOS and Android apps

LLM & AI: Anthropic Claude API, structured LLM output with validation and rule-based fallback, speech-to-text (faster-whisper), speaker verification (ONNX), computer vision (OpenCV)

Media: FFmpeg rendering and audio mixing, RNNoise, sidechain ducking

SOFTWARE ENGINEERING EXPERIENCE

Sole Developer — VidlyAI — vidlyai.com Oct 2024 – Present

- Designed and shipped an AI video editor that turns raw phone clips into captioned vertical reels, built on a Spring Boot/PostgreSQL API, a React web app, Capacitor iOS and Android apps, and a Python video worker. About 3,000 sign-ups.
- Decoupled the API and worker through a PostgreSQL job queue (FOR UPDATE SKIP LOCKED) with stale-job recovery; deploys run through GitHub Actions, gated on 331 engine tests, with automatic rollback.
- Stored each reel as an edit decision list so user edits re-render without re-running transcription or calling the AI, making edits free; cached render segments reuse unchanged work.
- Built a nine-stage pipeline: faster-whisper word-level transcription, OpenCV face tracking, and speaker detection combining mouth-movement scoring with a WeSpeaker voice fingerprint; Claude returns an edit decision list, validated against the source clips and rendered with FFmpeg.
- Replaced the audio cleanup chain with RNNoise neural denoising, raising the voice-to-background gap from 14.5 dB to 20.4 dB on a speech-in-crowd test clip.

Sole Developer — Eventful — useventful.com Jan 2026 – Present

- Launched a live multi-tenant community event calendar SaaS on Next.js 16, TypeScript, PostgreSQL/Prisma, and NextAuth v5.
- Built AI flyer scanning on the Anthropic Claude API that reads an uploaded image and auto-populates a structured event listing, replacing manual data entry.

Software Engineer — Field Day — ASU Capstone, team of 5 Jun 2024 – May 2025

- Built an offline-first Progressive Web App (React, Vite, Tailwind, Firebase/Firestore) for wildlife researchers running capture-mark-recapture studies in remote locations without connectivity. Sponsored by an ASU Biology professor.

PRIOR CAREER

- Owner and operator, motorcycle shop (2002–2020): started it with no formal training, ran it for 17 years, then sold it to return to school for software engineering.
- Clinical Engineer, Stanford / Lucile Packard Children's Hospital (1994–2001): worked on an infant nitric oxide delivery system and assembled and calibrated medical device hardware used in a clinical trial published in the New England Journal of Medicine. Later ran clinical engineering at Mount Zion Hospital as its sole clinical engineer.
- U.S. Army, Biomedical Equipment Specialist (35G), 1991–1994: first soldier to score 100% in biomedical equipment training, which led to an assignment at Walter Reed Army Medical Center. Two Army Achievement Medals.